

$$R_{nl}(r) = \sqrt{\left(\frac{2Z}{na_\mu}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-Zr/na_\mu} \left(\frac{2Zr}{na_\mu}\right)^l L_{n-l-1}^{(2l+1)}\left(\frac{2Zr}{na_\mu}\right)$$